

Git & GitHub Beginner Case Study

(Question-Based Lab)

Scenario: You have joined ABC Technologies as a junior developer. Your team is building a Student Management System using Git and GitHub. Complete the following tasks by answering the questions and performing the required Git operations.

1. Create a folder named **StudentManagementSystem**. Which command creates the folder? Which command initializes it as a Git repository?
2. Create README.md, index.html, css/style.css, and js/app.js. What does **git status** display after creating these files?
3. Stage all files. Which command stages every file? How does git status change?
4. Commit the staged files with a meaningful message. Why are meaningful commit messages important?
5. Display the commit history using both **git log** and **git log --oneline**. Which output is easier to read?
6. Update README.md by adding project modules. Which command shows the differences before committing?
7. Create and switch to a branch named **feature-registration**. How do you verify the active branch?
8. Add students/student.txt with registration details. Commit the changes. Why is feature development usually done in branches?
9. Switch back to the main/master branch and merge feature-registration. Did Git merge automatically? Why or why not?
10. Create a GitHub repository named StudentManagementSystem. Why should you avoid initializing it with a README when pushing an existing local repository?
11. Connect the local repository to GitHub. Which command verifies the configured remote repository?
12. Push your code to GitHub. What is the purpose of the -u option in git push?
13. Create another branch named feature-login, push it to GitHub, and create a Pull Request. Why are Pull Requests important in team collaboration?
14. Another developer has updated the repository. Which command retrieves the latest changes? What could happen if you do not pull before pushing?
15. You accidentally staged temp.txt. Which command removes it from the staging area without deleting the file?
16. Two developers modify the same line in README.md. What is a merge conflict? How do you resolve it?
17. Display the complete repository history using git log --oneline --graph --all. What information does this view provide?
18. Explain the difference between Git and GitHub in your own words.
19. List three best practices for writing commit messages.
20. Submit: GitHub repository URL, screenshots of commit history, branches, pull request, merge result, and final repository.